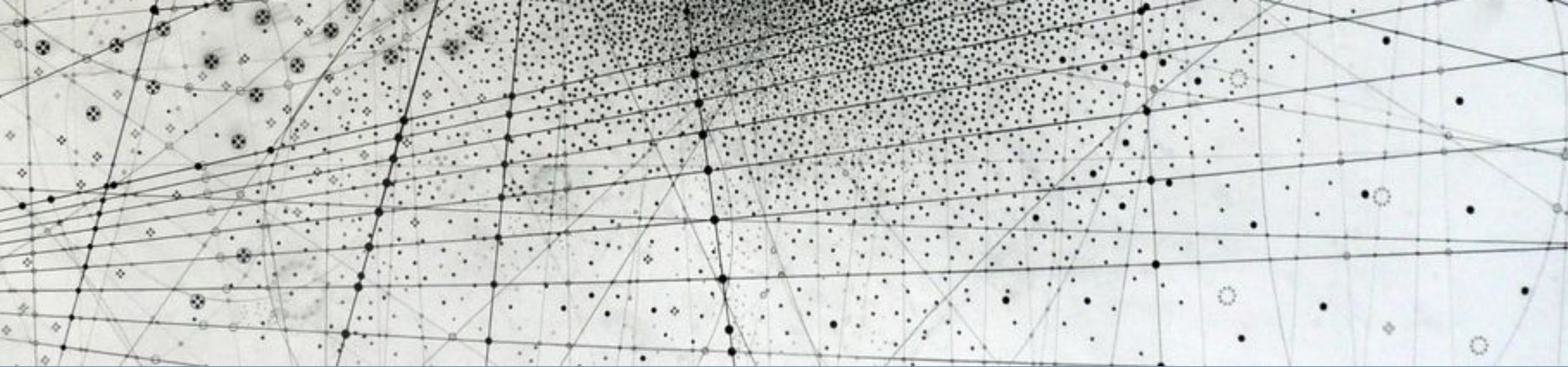


Research Methods in Machine Learning & Physics

Quick recap

Quick Recap: Geometric ML & Convolutions

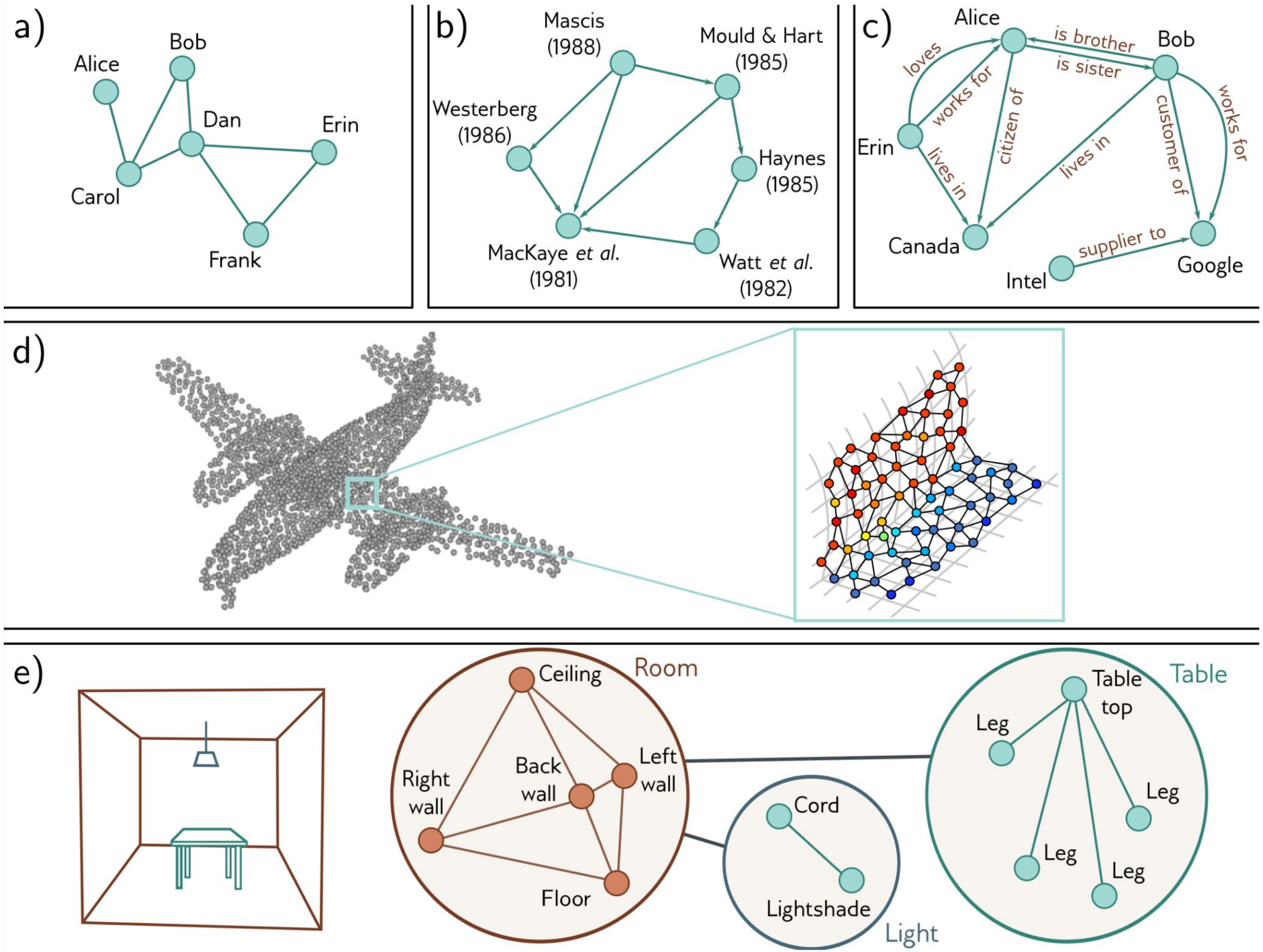
- **Geometric ML involves architectures designed to process certain data structures.**
  - These data structures generally have inherent *inductive biases*, e.g. images often have local 2D structure, sequences have a special ordering, and point clouds have neighbors in 3D space.

- **Invariance =** same answer regardless of an action of the symmetry group
- **Equivariance =** answers that vary according to the action of the symmetry group

- **Convolutional Neural Networks (CNNs)** process images via 2D convolutions.
- **Graph Neural Networks (GNNs)** process graph-structured data via graph convolutions.

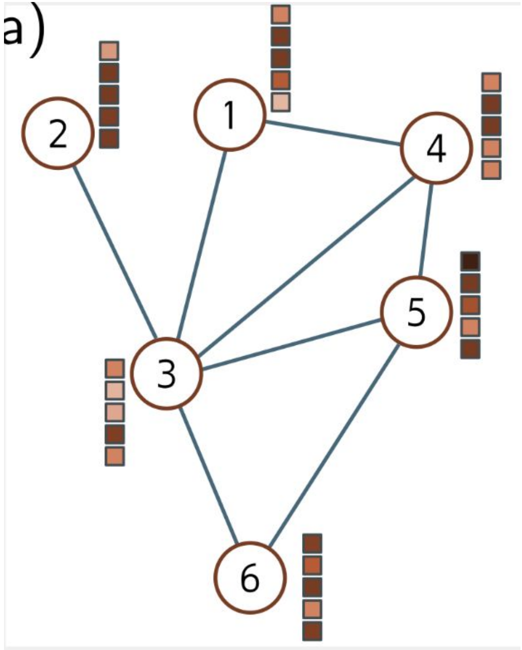
Graph Neural Networks (GNNs)

Graph data

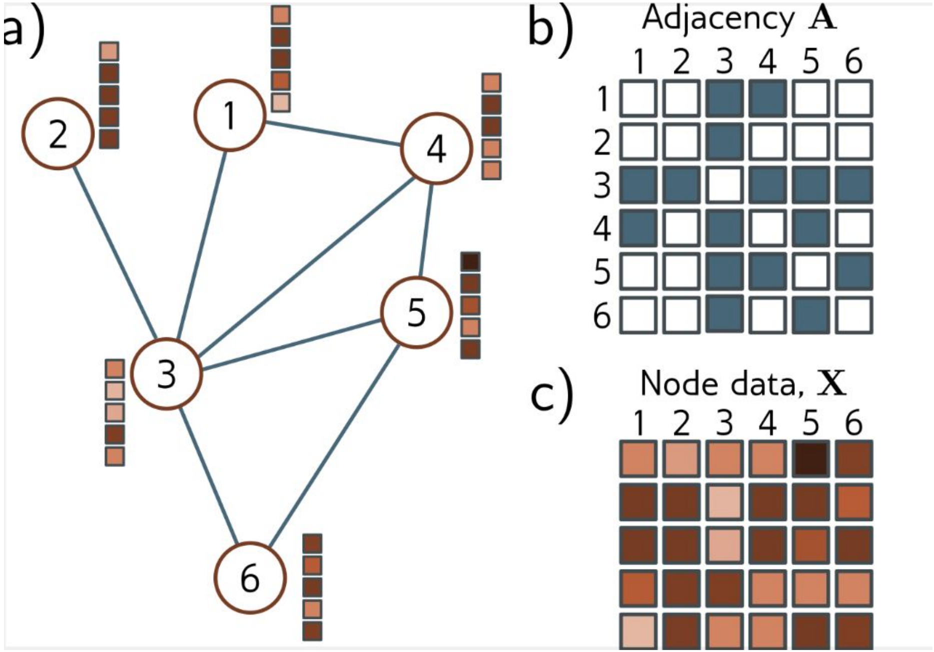


a)
Alice
Bob
Dan
Erin
Carol
Frank
b)
Mascis (1988)
Mould & Hart (1985)
Westerberg (1986)
Haynes (1985)
MacKaye et al. (1981)
Watt et al. (1982)
c)
Alice
Bob
is brother
is sister
loves
works for
citizen of
customer of
works for
Erin
lives in
lives in
Canada
Intel
supplier to
Google
d)
e)
Ceiling
Room
Right wall
Back wall
Left wall
Floor
Cord
Lightshade
Light
Table top
Table
Leg
Leg
Leg
Leg

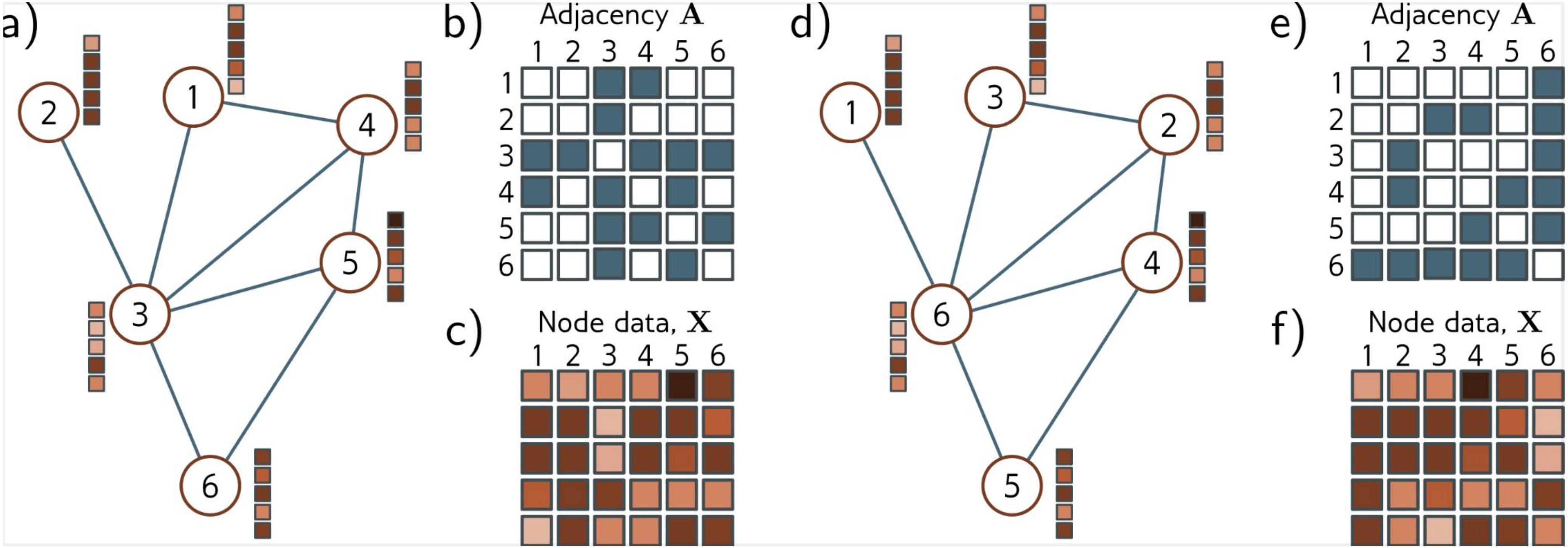
How to turn a graph into numbers?



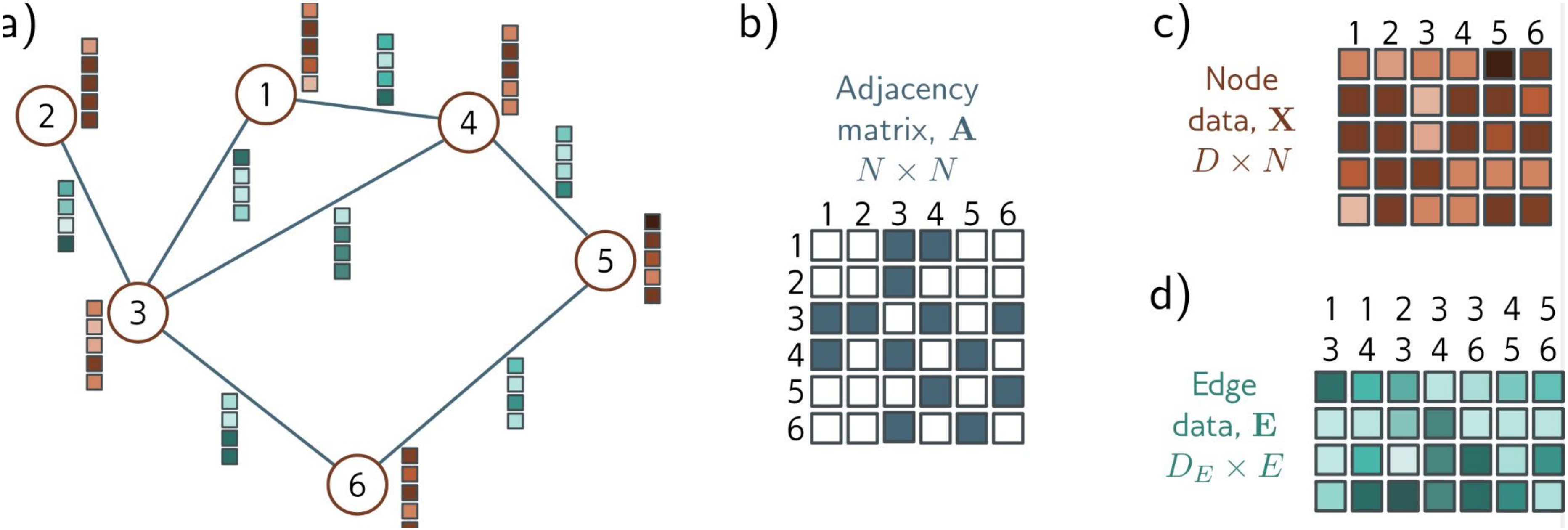
How to turn a graph into numbers?



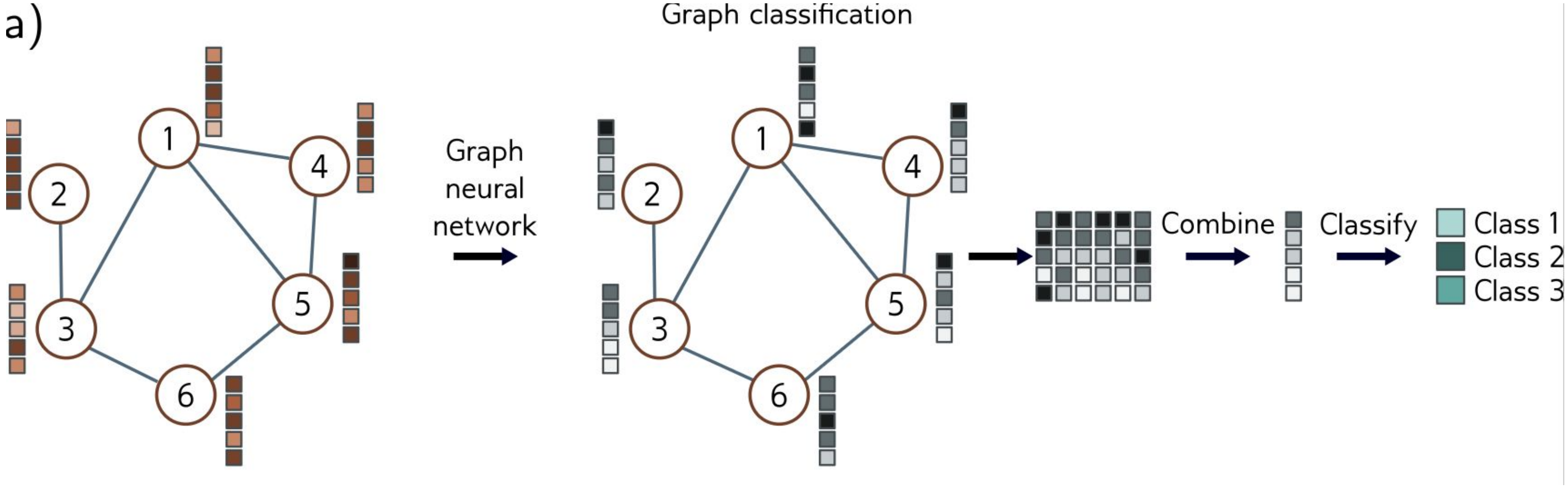
How to turn a graph into numbers?



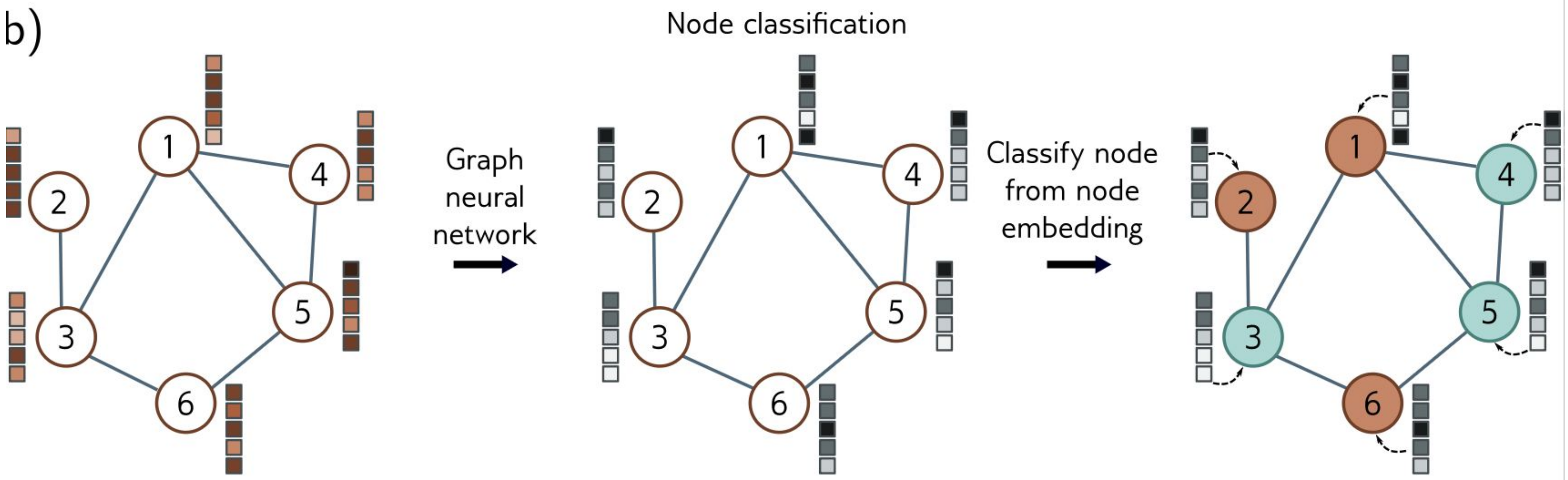
How to turn a graph into numbers?



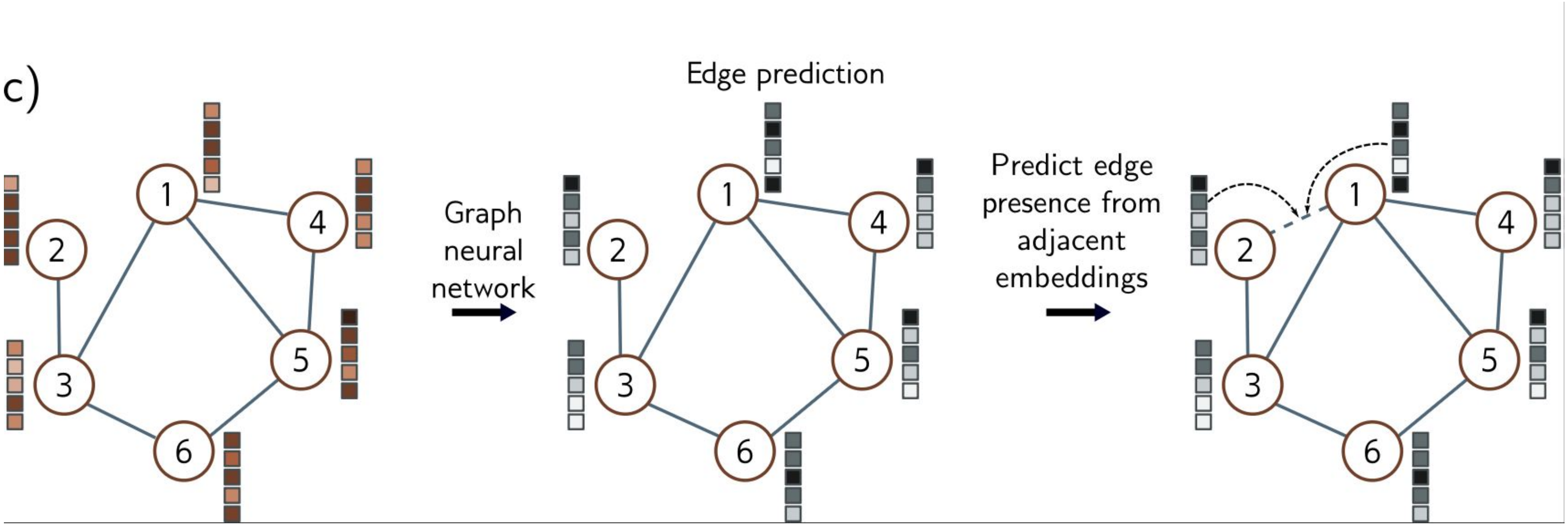
Possible tasks: **Graph classification**



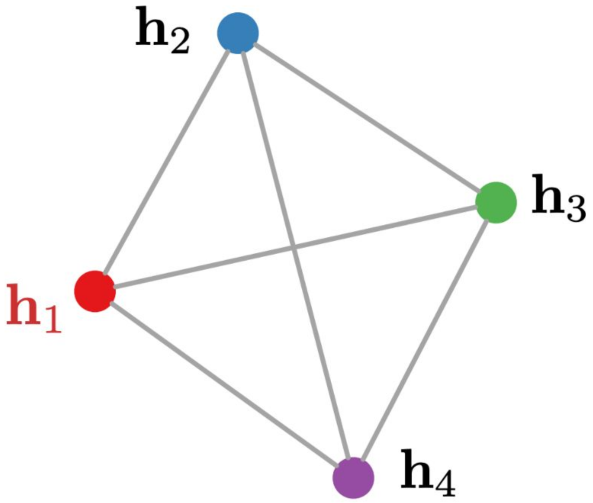
Possible tasks: **Node classification**



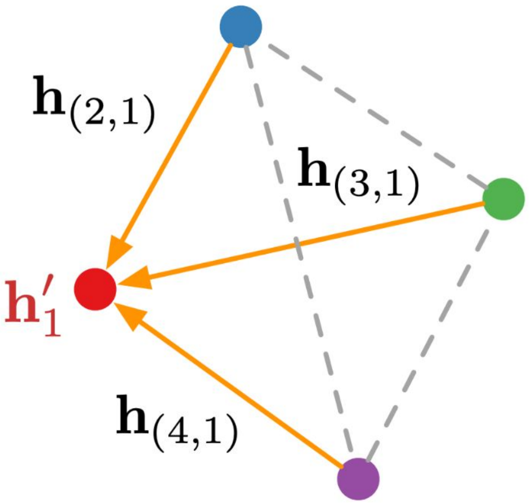
Possible tasks: **Edge prediction**



Graph Neural Networks



(a) Node representations.

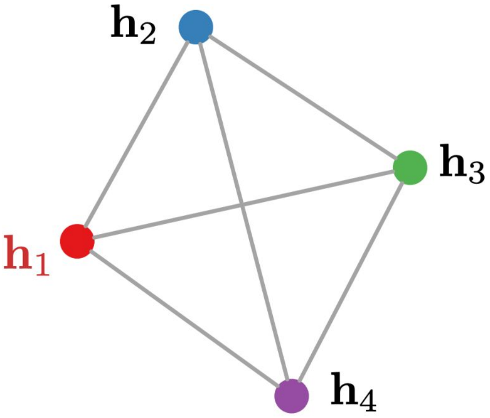


(b) Message passing step.

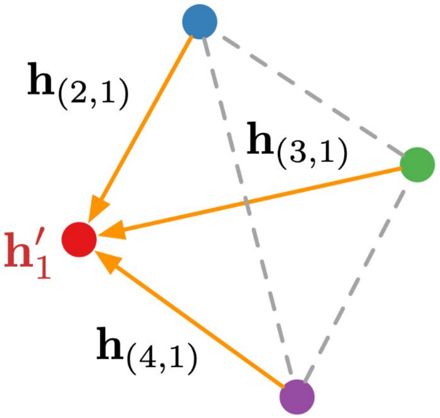
Graph Neural Networks

$$\mathbf{h}_{(i,j)} = f_{\text{edge}}(\mathbf{h}_i, \mathbf{h}_j, \mathbf{x}_{(i,j)}),$$

$$\mathbf{h}'_i = f_{\text{node}}(\mathbf{h}_i, \textstyle\sum_{j \in \mathcal{N}_i} \mathbf{h}_{(j,i)}, \mathbf{x}_i)$$

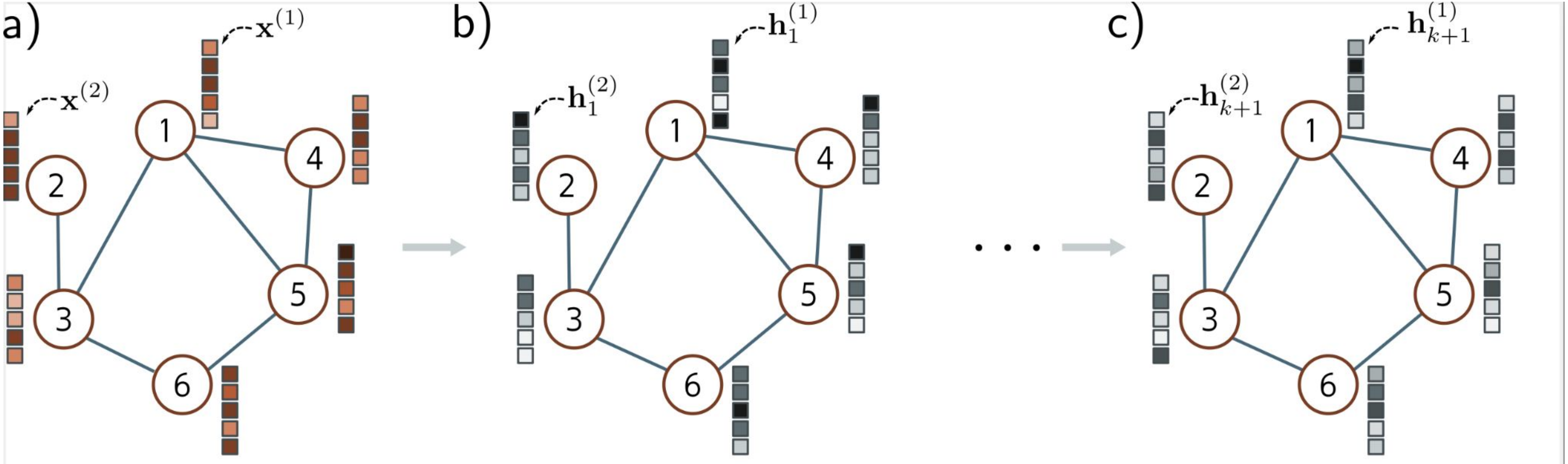


(a) Node representations.



(b) Message passing step.

Graph Convolutional Network (GCN)



$$\mathbf{h}_1^{(n)} = \mathbf{a}\left[\boldsymbol{\beta}_0 + \boldsymbol{\Omega}_0 \mathbf{x}_1^{(n)} + \boldsymbol{\Omega}_0 \mathbf{agg}[n]\right]$$

$$\mathbf{h}_{k+1}^{(n)} = \mathbf{a}\left[\boldsymbol{\beta}_k + \boldsymbol{\Omega}_k \mathbf{h}_k^{(n)} + \boldsymbol{\Omega}_k \mathbf{agg}[n]\right]$$

GRAPH NEURAL NETWORKS

FROM THE GROUND UP

In-class work

GNNs:

https://github.com/mariel-pettee/phys_805_fall_2025/tree/main/in_class_work

Name	Last commit message
..	
0_intro_notebooks_plotting.ipynb	update in-class work notebooks
1_intro_training_a_nn.ipynb	update in-class work notebooks
2_dimensionality_reduction.ipynb	Added in-class notebook on dimensionality reduction
3_graph_neural_network.ipynb	Added new in-class notebook for GNNs